

Architecture

Platform

- **MCU:** WCH CH32V003 family (RISC-V `riscv32imc` , QingKe V2 core as used with this target).
- **Toolchain:** `no_std` , no allocator; crates: `panic-halt` , `semihosting` (stdio for debug prints).
- **Assembly listings:** `cargo gen-asm` (release) and `cargo gen-asm-dev` (dev + `.loc`) emit `.s` files for studying RISC-V codegen — see [README.md](#).
- **Link layout:** Custom `memory.x` (see [memory.md](#)); build target `riscv32imc-unknown-none-elf` .

Module layout

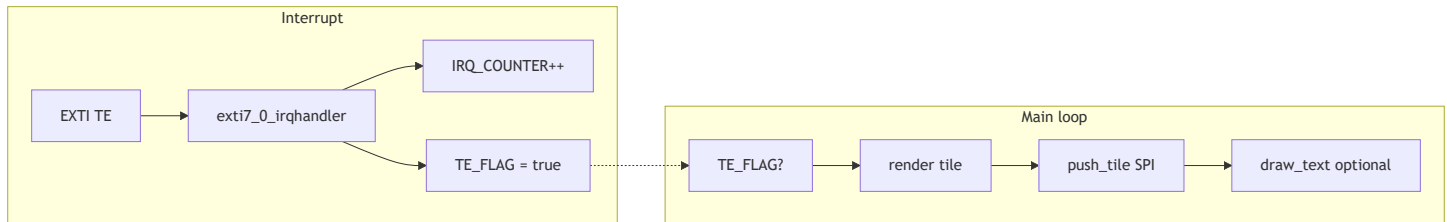
Module	Role
main	Entry, game loop, TE flag handling, tile render orchestration, IRQ counter + semihosting log
regs / rcc	WCH register map; 48 MHz HSE+PLL clock init
hw	GPIO (board pins), SPI1 @ 24 MHz , DMA1 ch3 TX , TIM1 CH1 PWM backlight on PD2 , delays
ili9341	C-style init table, DMA <code>fill_rectangle</code> / <code>push_solid_tile</code> , TE on
font	Minimal bitmap text (<code>draw_char</code> / <code>draw_text</code>) over SPI

There is **no** separate board-support package: `hw::platform_init()` (called from `ili9341::init`) configures RCC, GPIO, SPI1, and DMA. Backlight comes up at full brightness via `backlight_on()` → **TIM1** PWM on **PD2** (~47 kHz; duty `0..=1023` , max **1023** — see [hardware.md](#)). **EXTI/NVIC** for TE on **PD0** is still TODO in the IRQ handler.

Graphics pipeline

1. The display is treated as a **6×8 grid** of **40×40** RGB565 tiles (48 tiles total). `Tiles` holds a `dirty[48]` mask.

- When the **TE / EXT**I path runs, the main loop clears the flag, optionally logs `IRQ_COUNTER` via semihosting, picks the next dirty tile, computes a **solid RGB565 color**, and `push_solid_tile` (SPI **DMA**, no tile RAM buffer — same idea as the C driver's `ili9341_fill_rectangle`).
- A simple “**HELLO**” overlay uses `font::draw_text` , which rasterizes into a small static buffer and **DMA**-streams pixels.



Interrupts

- `exti7_0_irqhandler` matches the WCH naming for the **EXTI line 7–0** IRQ. On this board the panel **TE** signal is wired to **PD0** (see [hardware.md](#)).
- The handler only bumps a counter and sets `TE_FLAG` ; **EXTI pending clear** is left as a TODO in source (“platform specific”).

Debug output

- `semihosting::println!` sends text to the **host debugger** when RISC-V semihosting is enabled. It is **not** UART. If no debugger handles semihosting, output may be missing or behavior may depend on your probe.